ANSWER KEY & MARKING SCHEME SET 1

Chapter 3: **A Story of Numbers** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 22	Expand with powers of the base
2	**(c)** 101101	Repeated division by 2
3	**(c)** CXI	44+67=111
4	**(a)** 7	It is $111_2 = 2^3 - 1$
5	**(c)** 5	20 lies between 2^4 and 2^5
6	**(a)** 10001_2	11+6=17
7	**(b)** 100	Digit x (base)^(position from right, starting 0)
8	**(d)** 27	Expand with powers of the base
9	**(c)** 110110	Repeated division by 2
10	**(c)** CCXII	78+134=212

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$1 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$ (1); $= 53$ (1).	1+1
12	139 + 262 = 401 (1); CDI (1).	1+1
13	In decimal 26-9=17 (1); 10001_2 (1).	1+1
14	$2 \times 3^4 + 0 \times 3^3 + 1 \times 3^2 + 2 \times 3^1 + 1 \times 3^0$ (1); $= 178$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 45$ (1); $45 = 8 \times 5 + 5$; $5 = 8 \times 0 + 5$ (1); 55_8 (1).	1+1+1
16	Decimal values 5 and 3 (1); product 15 (1); 1111_2 (1).	1+1+1
17	1994 = MCMXCIV, 449 = CDXLIX (1); difference = 1545 (1); MDXLV (1).	1+1+1
	OR: $1 \times 5^3 + 4 \times 5^2 + 3 \times 5^1 + 2 \times 5^0 = 242$ (1); $242 = 8 \times 30 + 2$; $30 = 8 \times 3 + 6$; $3 = 8 \times 0 + 3$ (1); 362_8 (1).	1+1+1
18	Decimal values 6 and 5 (1); product 30 (1); 11110_2 (1).	1+1+1
	OR: 2024 = MMXXIV, 1999 = MCMXCIX (1); difference = 25 (1); XXV (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Add in binary: 11100_2 (3); decimal check $13 + 9 + 6 = 28$ (2).	3+2
	OR: Smallest $10000_2 = 16$ (2); largest $11111_2 = 31$ (2); count $32 - 16 = 16$ (1).	2+2+1
20	Add in binary: 100001_2 (3); decimal check $17 + 11 + 5 = 33$ (2).	3+2
	OR: Smallest $1000_3 = 27$ (2); largest $2222_3 = 80$ (2); count $81 - 27 = 54$ (1).	2+2+1

Mistakes to avoid:

- Repeating V, L or D, or writing IC, IL, XM for 99, 49, 990; use XCIX, XLIX and CMXC.
- Reading the remainders from the first to the last when converting decimal to another base; they must be read from the **last to the first**.
- Using a digit equal to or larger than the base (for example 135_5 is not a valid number).
- Taking the place values of binary as 1, 2, 3, 4 instead of 1, 2, 4, 8.
- Not carrying properly in binary: $1 + 1 = 10_2$, not 2.
- Confusing bases of ancient systems: Babylonian is base 60 and Maya is base 20, not base 10.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 3: **A Story of Numbers** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 26	It is $222_3 = 3^3 - 1$
2	**(a)** 6	50 lies between 2^5 and 2^6
3	**(c)** 10010_2	13+5=18
4	**(b)** 1536	Digit x (base)^(position from right, starting 0)
5	**(c)** 65	Expand with powers of the base
6	**(b)** 100101	Repeated division by 2
7	**(d)** CCCIV	96+208=304
8	**(c)** 124	It is $444_5 = 5^3 - 1$
9	**(b)** 7	127 lies between 2^6 and 2^7
10	**(c)** 10000_2	9+7=16

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	87 + 513 = 600 (1); DC (1).	1+1
12	In decimal 35-12=23 (1); 10111_2 (1).	1+1
13	$3 \times 5^3 + 4 \times 5^2 + 0 \times 5^1 + 2 \times 5^0$ (1); $= 477$ (1).	1+1
14	176 + 424 = 600 (1); DC (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$5 \times 8^2 + 7 \times 8^1 + 1 \times 8^0 = 377$ (1); $377 = 5 \times 75 + 2$; $75 = 5 \times 15 + 0$; $15 = 5 \times 3 + 0$; $3 = 5 \times 0 + 3$ (1); 3002_5 (1).	1+1+1
16	Decimal values 7 and 3 (1); product 21 (1); 10101_2 (1).	1+1+1
17	1486 = MCDLXXXVI, 349 = CCCXLIX (1); difference = 1137 (1); MCXXXVII (1).	1+1+1
	OR: $2 \times 3^3 + 2 \times 3^2 + 0 \times 3^1 + 1 \times 3^0 = 73$ (1); $73 = 5 \times 14 + 3$; $14 = 5 \times 2 + 4$; $2 = 5 \times 0 + 2$ (1); 243_5 (1).	1+1+1
18	Decimal values 9 and 5 (1); product 45 (1); 101101_2 (1).	1+1+1
	OR: 3000 = MMM, 1776 = MDCCLXXVI (1); difference = 1224 (1); MCCXXIV (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Add in binary: 101001_2 (3); decimal check $22 + 7 + 12 = 41$ (2).	3+2
	OR: Smallest $100_5 = 25$ (2); largest $444_5 = 124$ (2); count $125 - 25 = 100$ (1).	2+2+1
20	Add in binary: 110000_2 (3); decimal check $25 + 14 + 9 = 48$ (2).	3+2
	OR: Smallest $1000_4 = 64$ (2); largest $3333_4 = 255$ (2); count $256 - 64 = 192$ (1).	2+2+1

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- Not carrying properly in binary: $1 + 1 = 10_2$, not 2.
- Confusing bases of ancient systems: Babylonian is base 60 and Maya is base 20, not base 10.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 3: **A Story of Numbers** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 18	Digit x (base)^(position from right, starting 0)
2	**(d)** 241	Expand with powers of the base
3	**(c)** 111110	Repeated division by 2
4	**(a)** D	189+311=500
5	**(a)** 215	It is $555_6 = 6^3 - 1$
6	**(d)** 8	128 lies between 2^7 and 2^8
7	**(a)** 10001_2	14+3=17
8	**(b)** 24	Digit x (base)^(position from right, starting 0)
9	**(c)** 252	Expand with powers of the base
10	**(c)** 11101	Repeated division by 2

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	In decimal 47-18=29 (1); 11101_2 (1).	1+1
12	$5 \times 8^3 + 1 \times 8^2 + 6 \times 8^1 + 3 \times 8^0$ (1); $= 2675$ (1).	1+1
13	348 + 267 = 615 (1); DCXV (1).	1+1
14	In decimal 58-23=35 (1); 100011_2 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$1 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 = 51$ (1); $51 = 4 \times 12 + 3$; $12 = 4 \times 3 + 0$; $3 = 4 \times 0 + 3$ (1); 303_4 (1).	1+1+1
16	Decimal values 11 and 3 (1); product 33 (1); 100001_2 (1).	1+1+1
17	2450 = MMCDL, 899 = DCCCXCIX (1); difference = 1551 (1); MDLI (1).	1+1+1
	OR: $2 \times 6^2 + 4 \times 6^1 + 3 \times 6^0 = 99$ (1); $99 = 8 \times 12 + 3$; $12 = 8 \times 1 + 4$; $1 = 8 \times 0 + 1$ (1); 143_8 (1).	1+1+1
18	Decimal values 6 and 7 (1); product 42 (1); 101010_2 (1).	1+1+1
	OR: 1999 = MCMXCIX, 749 = DCCXLIX (1); difference = 1250 (1); MCCL (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Add in binary: 110000_2 (3); decimal check $30 + 11 + 7 = 48$ (2).	3+2
	OR: Smallest $100_6 = 36$ (2); largest $555_6 = 215$ (2); count $216 - 36 = 180$ (1).	2+2+1
20	Add in binary: 101010_2 (3); decimal check $19 + 15 + 8 = 42$ (2).	3+2
	OR: Smallest $100_7 = 49$ (2); largest $666_7 = 342$ (2); count $343 - 49 = 294$ (1).	2+2+1

Mistakes to avoid:

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